

Bitwise differential trail through **QALQAN** found by SAT

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1 QALQAN

Maximal differential probability: 2^{-42}

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	f	0	f	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	f	0	f	0
KayakStart																																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	f	0	f	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	f	0	f	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	f	0	f	0
QuayenRound																																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	1	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	1	
QuayenRound																																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	
QuayenRound																																	
0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0	8	0	0	0	0	8	0	8	0							
0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0	8	0	0	0	0	8	0	8	0							

2 QalqanRound

Maximal differential probability: $2^{-14.0}$

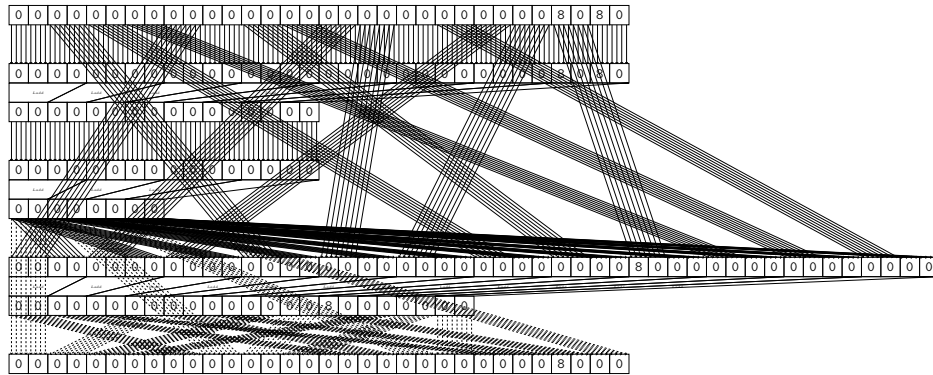
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	f	0	f	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	f	0	f	0
8 Blue Layers																											
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	8	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	8	0
L																											
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0
Add 126																											
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	1
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	1

3 SBoxLayer

Maximal differential probability: $2^{-12.0}$ [illegible]

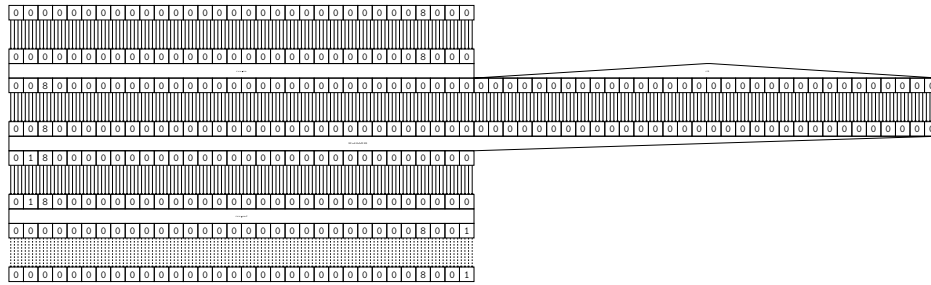
4 L

Maximal differential probability: $2^{-0.0}$



5 Add128

Maximal differential probability: $2^{-2.0}$



6 QalqanRound

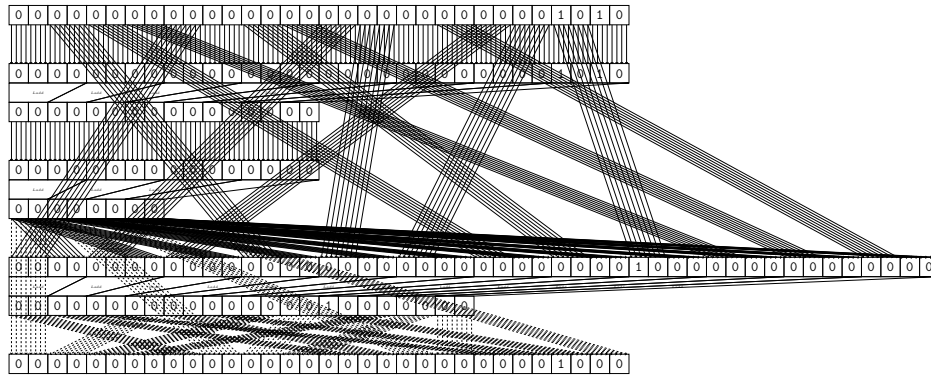
Maximal differential probability: $2^{-17.0}$ [illegible]

7 SBoxLayer

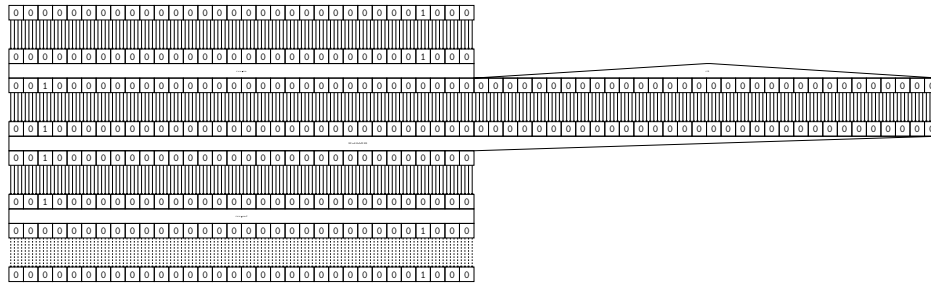
Maximal differential probability: $2^{-14.0}$ [illegible]

8 L

Maximal differential probability: $2^{-2.0}$



9 Add128

Maximal differential probability: $2^{-1.0}$ 

10 QalqanRound

Maximal differential probability: $2^{-11.0}$ [illegible]

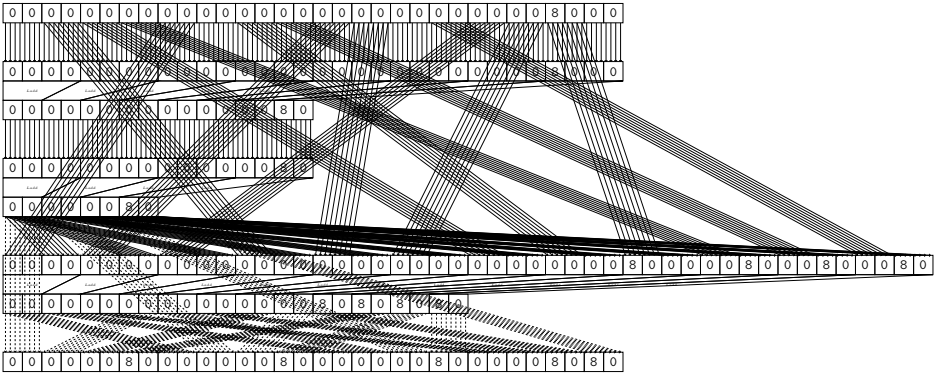
11 SBoxLayer

Maximal differential probability: $2^{-7.0}$

[illegible]

12 L

Maximal differential probability: $2^{-0.0}$



13 Add128

Maximal differential probability: $2^{-4.0}$

